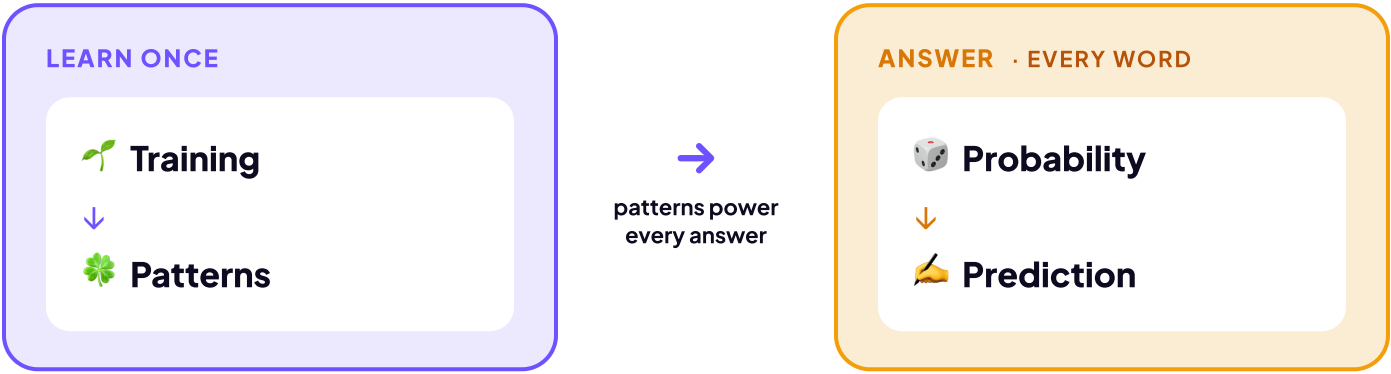


How We Got Here

AI can feel like it appeared out of nowhere, fully formed, almost like magic. It didn't. What looks like one sudden invention is really four older ideas, each worked out by different people who weren't trying to build AI at all.

You already met those four ideas in How an LLM Works. Here's how they come together: four separate breakthroughs that finally connected at the single moment everything changed.



None of them arrived together. Each was figured out decades, even centuries, apart, for reasons that had nothing to do with AI.

Four ideas, invented separately, that finally connected

**Probability** 1650s

Started as gambling math: a way to figure the odds. It lets a model answer with how likely something is, never a flat certainty.

PASCAL & FERMAT

**Prediction** 1948

Treat language as guessing the next word from the words before it, the same idea behind your phone's autocomplete.

CLAUDE SHANNON, A MATHEMATICAL THEORY OF COMMUNICATION

**Training** 1957

The first machine that could learn patterns from examples instead of following hand-written instructions.

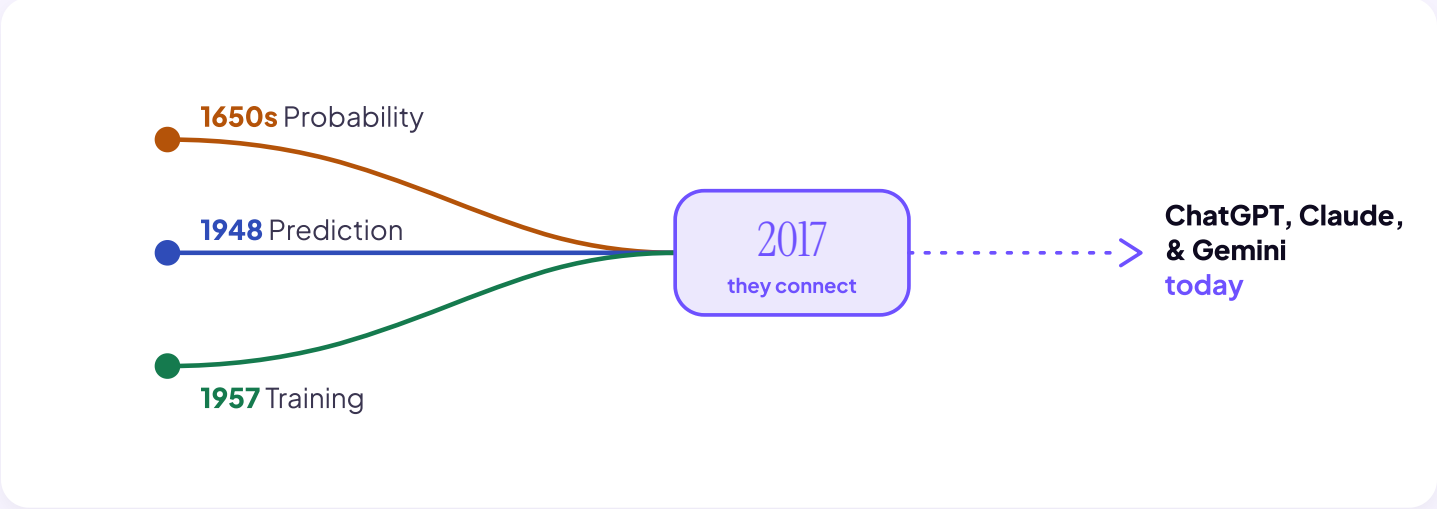
FRANK ROSENBLATT, THE PERCEPTRON



**Patterns** 2017

Eight researchers at Google published one breakthrough that let a machine weigh all the words at once and find the patterns between them: the transformer, the spark that finally connected the other three.

VASWANI AND COLLEAGUES, ATTENTION IS ALL YOU NEED



 **Centuries apart, then all at once.** Four ideas, each invented separately and left sitting apart for decades, finally snapped together in 2017. That collision, not any single invention, is the AI you use today.